MALMÖ UNIVERSITY

Reimagining Kitchen Artifacts through Future Catastrophes

A speculative and critical design critique on the present values in kitchen design

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Spring 2025
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Abstract

Why should we speculate about the future through design? Why should we care about kitchens and food? This thesis unfolds these questions by researching the field of HFI and the issue of efficiency in kitchen artifacts design, by looking at future crises. In the age of the Anthropocene, we are facing more challenges than opportunities, which design should address - offering alternatives. I propose 2 scenarios, “back to basics with food” and “techno-centric kitchen”, that I use to encourage people to think about the future of food through collaborative brainstorming workshops. The results of the workshops and fieldwork led to the creation of 2 speculative artifacts embodying these futures: the Preservation Scale and the Toastless Toaster. When testing the legibility of the message within them, most people reflected on their disconnection with food. With this project I endeavour to enrich the speculative field of HFI by proposing an approach to the black-box metaphor and imitating a feeling of its mastery, and also offer future research opportunities within this niche.

Acknowledgments

I want to express my gratitude towards everyone who helped me through this journey. Thank you to my supervisor Elisabet for her continuous support in this journey, to my closest friends Filippo, Eszter and Marvin for being supportive even when it was difficult, to all the people that participated in this study, and to Marcus and Doro for inviting me to discuss around the topics surrounding this study.

I also want to thank my family for being by my side and supporting me in my search for knowledge.

Thank you!

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1 Introduction

The turning point of human evolution was the domestication of fire. Adopting the very thing that all nature fears, not only put humans on top of the food pyramid but also scaffolded evolutionary traits that altered the body to what we are today. But what did the adoption of fire mean? One of the most important benefits of fire was the ability to cook food (Wrangham, 2009). Cooking raw meat allowed for its safe and easy consumption, which in turn secured the nutrition for the body, and most importantly the brain (Wrangham & Carmody, 2010). Changing the diet, altered many aspects of the human anatomy, as well as how the life of the humanoid looked like, as fire was used for pottery beyond cooking. With this short trip back in time, I wanted to highlight two important things. First, through change of the primitive diet and consumption of enough nutrition, humans managed to evolve and in turn nurture their cognition. And second, the humans approached the very thing that nature fears, hence raising the question: What made humans approach the fire, the cornucopia of opportunities?

Nowadays, instead of fire, we rely on technology not only to cook our food but also to produce, transport, and commercialize ingredients—from agriculture to logistics and retail. Nevertheless, we are very much involved with food, as it holds an important cultural significance for people, beyond consumption. As such, many scientific fields have started researching food and its relation to people. Scholars within the field of Human-Computer Interaction have also exhibited interest in this domain, hence the rise of the field of Human-Food Interaction (HFI). Within the field of HFI, scholars explore ways of enhancing and improving interaction with food on all levels, from production to consumption. As a field, HFI observes the rise of certain trends in terms of research and development. In their taxonomy of the field, Bertran et al. (2019) present the different categories of projects within the field, as well the insights that arise from this categorization. However, it has been recognised that the field of HFI predominates with projects related to the trend of efficiency (Bell & Kaye, 2002; Bertran et al., 2019; Dolejsova, 2018). In other words, there is not much focus on improving the experience of people in their interaction with food, beyond being efficient. Moreover, Bertran et al. (2019) highlight the lack of research situated in the speculative category within HFI, which concerns projects that challenge the current trends. Similarly, Dolejsova (2018) emphasises the scarcity of speculative design methodology in food-oriented research.

The purpose of this thesis is to challenge the trend of efficiency in HFI, through Speculative and Critical Design (SCD), with background of Interaction Design (IxD). I approach the issue of efficiency by developing speculative kitchen artifacts, inspired by future scenarios and fieldwork insights. The future scenarios are anchored in societal issues that we are

beginning to face and in the near future will be inevitable. A case in point is food security, as one of the most unsettling issues we will face. We experienced a fraction of it during the COVID-19 pandemic, when logistics and exportation were limited, hence reducing the availability of certain products. Nevertheless, during that time people had employed different coping mechanisms, such as baking bread or growing vegetables at home.

I focus on SCD, because it has the potential to challenge the current assumptions that technology is always good and can solve all problems, while also speculating on the future wicked problems (Dunne & Raby, 2014). As it will become evident in the paper, current domestic technology is designed to standardise all experiences, especially when it comes to food practices.

It is also important to note why I began this paper by examining the early humans' adoption of fire. Instinct pushes people away from uncomfortable things. But then the question is why did early humans approach the fire, from which all animals fret? In their paper, Young (2022) argues that human creativity was the driving force behind the invention of cooking and the creation of language. It is without doubt that after the fire was adopted humans became even more cognitive and creative, because of the endless possibilities, from cooking to ceramics. The point is, this encounter with the uncomfortable brought humans the evolutionary advantage. In this day and age, we need to prepare for the uncertain future to come, and one way to achieve this is to face the discomfort and approach challenges together.

This thesis follows the 12-week project and showcases the different phases of the design process. I begin the exploration with an in-depth theoretical investigation of the status quo in designing for the home and narrowing it down to the kitchen and related work. Then, I examine SCD and the rest of the methods that guided the project. The next chapter showcases the practical part of the project, going through the fieldwork and the materialisation of the concept. Finally, I will elaborate on the most important findings in the discussion and suggest future opportunities within this area of study.

1.1 Research questions

This project and analysis are led by the main research question (RQ):

How might we challenge the trend of efficiency in human-food interaction design, through speculative and critical design?

Within this general question I locate two sub-questions to assist with answering the main question:

1. How might we prepare people for the ambiguous future surrounding food practices through SCD?
2. How might we critique the current mediation of technology in the kitchen through SCD?

1.2 Contributions

The contributions of this project lie within enriching the underdeveloped domain of speculative design within the field of HFI, as well as locating future opportunities for food-oriented SCD research.

1.3 Limitations

This thesis explores the broad topic of speculative design within HFI and more specifically interaction with kitchen artifacts. The research deals with problematising kitchen design's focus on efficiency and the societal problem of food security through future scenarios and artifacts based on these futures. The prototypes were tested with a narrow testing group, implying the need of further study.

2 Background

To set the scene, this chapter will introduce what is already known in designing for the home and the current status quo in the field of HFI. I begin by introducing the home as a design space and how it has been done so far, which then narrows down to the home as an ecosystem and the things that design scarcely considers. The next section talks about the kitchen as a centre of the house and the focus of this project. The last section covers related work within SCD and food practices.

2.1 Design for the home

Ever since computers were introduced to the office, there has been a surge of curiosity around their use. Eventually, they penetrated the walls of the home and became embedded into the home ecosystem. Thus, designers shifted their focus from the office on designing for the home. However, one of the flaws that followed was that designers used the same rationalised methods for designing for the office when designing for the home (Bell & Kaye, 2002; Crabtree & Rodden, 2004). This in turn influenced some kitchen technology to be designed with the idea of production and efficiency (Crabtree & Rodden, 2004). This issue stretches to every room in the home, especially the kitchen, as it will be covered in the next sections. In the following subsections, I will cover the issues of design shifting from the office to the home and after that explore the issue of technological solutionism, which plagues corporations creating technology for the home.

2.1.1 From the office to the home

As mentioned before, the adoption of the computer in the home introduced the home as a design space. Gaver (2002), in Designing for Homo Ludens,

mentions the revolution of computing, transcending from its task-oriented background to our everyday surroundings, allowing us to get work done and do the things we enjoy. However, Gaver (2002) also highlights an important issue that as computing was conceived in the rational space of the office it has also inherited the values of the workplace, bringing them into the home once adopted.

The next important concept to be considered is the Homo Ludens, or "Man the Player", a term introduced by Huizinga (1938), nearly a century ago to highlight the importance of play in people's lives. The main critique in the book is the fact that civilizations are growing more serious, forgetting the concept of play. Gaver (2002) emphasizes on the characteristics of Homo Ludens and how we could design for them. The most important point that the author makes is the need to place pleasure before performance. As such, technology should aid people in their seek of pleasure, but most importantly encourage them towards new means of pleasure (Gaver, 2002). In this way, design supports the playful characteristics of people, such as curiosity, creativity and emotion, stemming away from the rational and logical nature of computing.

At the same time Gaver (2002) also suggests that designers should also be ready to recognise that they are Homo Ludens as well, implying the influence of the designer's play on their ideas. Recognising that we are also playful is vital to design, in order to produce something that is based in human nature and not just a superficial endeavour to bring computers to every aspect of our life. SCD has the potential to help designers escape the boundaries of normative design and embrace the Homo Ludens approach.

2.1.2 Technological solutionism

The search for better interaction and more pleasure has encouraged an immense evolution in computing throughout the span of a few decades. Concepts such as the smart home or the home of the future, have been a driving force for designers and entrepreneurs with focus on domestic technology. This however, has pushed an agenda on most corporations, specifically in the Silicon Valley, of improving everything from politics to cooking, as critiqued by Morozov (2013). What Morozov (2013) defines as "technological solutionism", refers to the overall simplification of social problems, which in turn results in the creation of superficial solutions, searching for a problem to solve. What is even more critical than the solution, as recognized by Morozov (2013), is the way we define the problem, equally vital for design practitioners.

The phenomenon of "smart homes" has been critiqued by many researchers, all of which suggest what the long-term consequences of such endeavours would be like (Bell et al., 2005; Bell & Kaye, 2002; Dolejsova, 2018; Morozov, 2013). Bell and Kaye (2002) even compare the modern house as a "showroom" for all new gadgets, depriving the space from becoming a home.

Similarly to Morozov (2013), Bell and Kaye (2002, p. 53) too critique tech companies for viewing the domestic space as a "potential site for the consumption of computing power and the use of the Internet.".

To produce meaningful design, means having a profound understanding of the problem. It is not enough to just wish to improve people's lives, as there are multiple aspects to take into account when designing for the home, one such being recognizing the home as an ecosystem (Bell & Kaye, 2002).

2.2 Home as an ecosystem

To design for homes, means understanding the practices of the inhabitants within the home. As such, it is important to consider how they utilize their space, the way they use technology and most of all consider how their culture influences their choices. Technology for the home should be designed in a way that it does not obstruct or diminish any naturally occurring processes. Crabtree et al. (2012) highlight the importance of ethnographically informed design, especially when it comes to design for the home. Taking part in the practices done in the home, allows designers to grasp the understanding of the household members. In this section, I will discuss the importance of understanding the culture and routines in the home, as well as how people may shape the technology in the house, based on the former.

2.2.1 Culture and routines

"… routines are the very glue of everyday life" is how Tolmie et al. (2002, p. 399) describes the everyday activities that allow for the organization and smooth operation within the house. Routines vary from household to household, but they will always be found if the practices are scrutinised. It is also important to consider that routines will change throughout time, which means that routines in the past are totally different from what we know today or what we will see in the future, hence design should be mindful of that. Similarly, in the context of skill, Sutton (2006) highlights that new technology develops new gestures, on the expense of old ones.

Developing a skill means that the agent copes with a situation, with little to no consciousness of their actions, or as Dreyfus (2002) phrases it "a mental representation". There is a similar phenomenon in the critique of routines by Tolmie et al. (2002), as he describes that routines are invisible for those doing them. In a sense we can look at routines as skills, in keeping the home running smoothly. Technology also comes into the picture, as it may or may not support these routines. If it is the latter, then we can observe how the people configure the technology and the environment around their activities (Rodden and Benford, 2003, as cited in Crabtree & Rodden, 2004).

2.2.2 Humans shaping technology and vice versa

To design for skill and routines, means to design for coping. Coping is the way we go through actions and answer adequately to any resistance that may

occur (Heyer, 2018). It is heavily embodied, which means that there is no mental representation of the action but instead, it is the body that guides us through the task (Dreyfus, 2002). In many situations, we use artifacts to achieve the desired outcome. By operating skilfully with the artifacts, they become an extension of the body (Heyer, 2018; Sutton, 2006). In this framework Heyer (2018) examines two lenses in relation to artifact design: malleability and manipulability. To elaborate on Crabtree and Rodden (2004), configuring the technology and the spaces around the house can be related to coping theory, especially in terms of malleability. Sutton (2006) provides a similar example of a participant adapting an everyday object to a specific cooking artifact that cannot be found outside the traditional home.

On the other hand, we can also think about how technology shapes the people. It is essential to consider what designers bring into the design, in terms of knowledge, because at the end of day design is politics. DiSalvo (2021) highlights the notion of design to produce culture. When we design systems we enter the process with certain knowledge, prejudices and most of all certain agendas. To elaborate on this, our knowledge is situated and limited to our background, and as Suchman (2002, p. 96) highlights, this fact should be used as an impetus for designers to reflect and feel responsible that "...our vision of the world is a vision from somewhere". They also stress the importance of locating design to its sites of use, which essentially means going out of the lab into the field. The feminist critique of "design from nowhere" (Suchman, 2002) is primarily targeted at design systems that are produced for the sole purpose of commercialization. As such, they are produced in a generalised manner, to accommodate every customer. This critique can be linked to the issue of technological solutionism and standardization.

With the politics of design, it is important to not only consider how the user may modify the design to their needs, but also understand that we enter the design space with a certain agenda. Hence, it is vital to base the design in its site of use. Commercial and "ubiquitous" design essentially diminishes culture and in the setting of the home it remains an "alien" that cannot adapt to the ecosystem. For this current project, the focus is on the kitchen, which the next section will explore.

2.3 The kitchen as the centre

The nucleus of the home, this is the kitchen. In many traditional homes, it is considered as the heart of the home, where people rest and eat together – a site for social connectedness. However, the kitchen has observed many changes in the way people look at it. Eijnde (2020) presents a short trip through the history of the kitchen, and how it evolved from the heart of the home and gradually separated from the common room to an isolated one for cooking. Alexander et al. (1977, p. 661) highlights this shift by naming the kitchen "an efficient but unpleasant factory for food". Many other scholars

have recognised the growing body of research done with the focus on efficiency in food practices, neglecting the importance of enhancing the experience (Bell et al., 2005; Bell & Kaye, 2002; Bertran et al., 2019; Dolejsova, 2018). This part of the background will deal with exploring the current challenges and gaps within the field of HFI, as the basis of the design work.

2.3.1 The problem of efficiency

So how did efficiency become so prominent in kitchen design? As mentioned above, the kitchen has gradually moved from the centre of the home to the periphery. However, this phenomenon differs from culture to culture. Bell et al. (2005) examines the history of American homes, with emphasis on how the industrialization and consumerism culture have influenced the design of domestic kitchen technology. With industrialization there was also a great focus on science as a way to measure all aspects of life, including kitchen and food practices (Bell et al., 2005). One of the most important consequences of this blind trust was the standardization of things. Bell et al. (2005) and Sutton (2006) both critique the standardization of food and cooking resulting in ethnic cooking being seen as unhealthy and family recipes being submerged by all science-approved methods and measurements. This is a prime example of design from nowhere, as practices from the lab have been standardized for all people.

The trend of efficiency is problematic, not only because it puts emphasis on the industrial, but also because it generalises people's needs. Bell et al. (2005) has recognised this as a design challenge, that not all users are the same, hence their needs vary. To work around this challenge, it is important to take on a user-centred design framework in order to be inspired by the processes in the home, and the kitchen. The transition of design in the workplace to the home, has further nourished the problem of efficiency, as implied before. Many technology systems are designed outside of the home, or are adopted directly from the office, such as the computer, hence the focus on efficiency. Bell and Kaye (2002) highlight the television, as an exception of a technology that has been developed in the home, hence is accepted and integrated well. This further illustrates the importance of locating design in its place of use, and not designing from the lab.

The complex history behind this phenomenon and the multiple factors influencing it showcase the complexity of the issue. It is also vital to recognise that efficiency as a problem has multiple aspects. The next sections will cover the most important ones that concern this study.

2.3.2 Agency in cooking

When discussing the idea of future smart technologies in cooking, most people in the study from Xu et al. (2023) focused on automation as a main aspect. While this study is limited to Chinese college students as the focus

group, it is worth noting that the younger generation perceives automation as the future in the kitchen. Nevertheless, as the authors also note, the participants expressed a strong need to be able to take control over the automated process (Xu et al., 2023).

Automation is closely related to efficiency, and has been recognised as a problem in itself, mainly as it endeavours to substitute the human factor in cooking. Bertran et al. (2019, p. 3) highlights this insight, as most research gives "greater autonomy to technology rather than the person". Similarly, Sutton (2006) describes the role of the woman in the 1950's kitchen as an assembler and not a cook. This is problematic as it sharpens the argument of the kitchen being a factory, filled with machines and lack of humans. Furthermore, relying on technology to execute everything in the kitchen means that we are also being de-skilled (Short, 2003). Deskilling, as Short (2003, p. 14) describes it, is a result of "technological rationalising systems", arising from the industrial idea of simplifying the whole process. Deskilling also occurs because of the purchase of pre-made and convenience foods (Short, 2003), resulting in loss of nutritional knowledge (Sutton, 2006) and sharpening the gap between people and their food.

One of the main issues with the lack of human agency in cooking is the fact that creativity is damaged. In their paper, Grimes and Harper (2008) highlight the issue of "corrective technology", which is designed to fix assumed problems during cooking. In other words, the corrective technology is trying to standardize human behaviour during cooking. The authors suggest the concept of "celebratory technology", which celebrates people's interaction with foods, in all aspects (Grimes & Harper, 2008). This not only supports people in their interaction, but also does not hinder experimentation and creativity, encouraging play. One of the main critiques for automation is that it follows a straightforward path, in order to minimize the "mistakes" the user could encounter. Grimes and Harper (2008) emphasise that mistakes for the system could be the exploration of the user while cooking, which in turn makes the cooking more personal, for example by creating your own signature dish. Smart technology, as Morozov (2013) underlines, as it gets smarter leaves no room for interpretation, resulting in higher standard food, but almost no pleasure in the process.

It is important to remember that by automating the cooking experience, humans are also moved out of the kitchen. The result of this is a gap between people and their food production and food source, a kind of isolation.

2.3.3 Isolation

When designing for the domestic, it is important to remember that the home is part of a community. Bell et al. (2005) highlights feminist theory by Hardyment (1990), that concerns the domestication of technology as a "servant", which in turn actually isolates the household members from the community. In their paper Bell et al. (2005) examine online shopping as an

example, where the household members can choose and order their products, with no interaction with any humans in their community. Similarly, in an earlier paper, Bell and Kaye (2002) present an ethnographic study in France, and highlight the importance for French people to go out to the market to buy products, because they can interact with the food and the people. Having this opportunity to go out and interact with the community allows people to feel connected and not isolated. Bell et al. (2005) highlights this design challenge, because design often looks at the home, without considering its place within the community, which results in making the home an "island", separated from the community.

Another infamous perpetrator of isolation is the supermarket, as recognised by Cerpina (2022). The supermarket has an abundance of food that we can access at all times. The problem with this is that it alienates people from the place the food is obtained (Cerpina, 2022). We can refer to the times when we adopted the fire and began hunting our food, and compare to the current times where all we need to do is go to the supermarket, without even questioning the source of the food. This is an important point to make, as food is also culture which connects people in the community at the end of the day. Bell and Kaye (2002) highlight the importance of food as a way to reflect one's role in the community, as exemplified by their study in Italy. In particular, the authors describe that the food served at the family's table is sourced from different places, which allows the cook to connect with the community and bring meaning to their food.

Isolation between people and their food has been recognised as problematic, for multiple reasons that I stated above. It is important to consider how such a phenomenon may influence the future of mankind, for food is vital for our survival.

2.3.4 The Ambiguous Future

Humanity has observed many catastrophes throughout its existence, and currently we observe the beginning of many others to come. A case in point is environmental disasters that may affect wildlife, and consequently food sourcing. Surviving in this ambiguous and catastrophic future, means learning how to adapt to the changes we cast upon our environment. As Cerpina (2022) highlights in their book, the survival of mankind depends on the ability to source and preserve food. With the current focus on automation and convenience, we essentially deprive ourselves from this knowledge, leaving us empty-handed to battle the future. Cerpina (2022) emphasises the importance of food, by bringing light to examples of societies that have met their demise, because of their inability to adapt to the changes they bring. "The progress trap", as the author call it, is the advances that initially seem beneficial to society, but in the long run bring disasters. The example in the book is of hunting, which eventually drained the resources for survival (Cerpina, 2022). Similarly, Feenberg (2002, as cited in Dunne & Raby, 2014)

expresses that even if we are aware of the technological potential, we still lack the appropriate intellectual means and tools to manage it. Hence, it is important to consider how designing domestic technology affects people and their survival skills.

One of the most feared catastrophes is food security. Cerpina (2022) also draws attention to this issue, as even minor disruptions could affect the supply chain and the food that we have access to, such as during Covid-19. Furthermore, such a catastrophe as Covid-19, is bound to alter our eating habits (Cerpina, 2022; Pitstick, 2021). It is during this time when we experience "breakdown", as in the case of coping (Heyer, 2018). The lack of certain products makes us stop and think of how to source it in other ways, or even substitute. It was the catastrophe that encouraged people to take on various domestic hobbies related to food, in order to meet their needs (Pitstick, 2021). In particular, baking has been recognised as a coping mechanism both for food security and a stress-relieving experience (Pitstick, 2021). As Pitstick (2021) further mentions, baking is a time consuming activity which makes the baker feel in control over what they do, a critique that goes against the purpose of efficiency-based technology.

Related to food security is the loss of skills, because of automation. Skills in terms of food sourcing and preservation are in risk of perishing. With the advent of smart technology, standardized food practices and automation we enter the space of the lab. Corrective technology (Grimes & Harper, 2008) and technological solutionism (Morozov, 2013) concern this phenomenon, as both strive for perfection and mistake prevention. Nevertheless, as the authors argue, cooking is based on trial and error, which is to be celebrated and not corrected. Furthermore, Morozov (2013) critiques the current vision of augmented technology, which appears to be intellectually deprived. The point in this critique is that through augmentation, instead of trying to impose efficiency, essentially making cooking less demanding, it is about making cooking challenging, in order to escape the monotony of cooking practices. Morozov (2013) elaborates that through the rationalising nature of efficiency, technology essentially prevents humans from achieving the skills in what they are doing. By making cooking more challenging, technology can bring more pleasure and encourage experimentation during cooking, enhancing creativity, similarly to the critique of Homo Ludens (Gaver, 2002).

When we speak about skill we turn back to the "mental representation" of Marley Ponty. An interesting perspective is proposed by Sutton (2006), who speaks of skills in relation to memory, especially in terms of recipes. For them, recipes are memory-jogs, which guide the cook through the skills they already possess and apply for different recipes. What is even more interesting is the idea of objects being an extension of the body allowing for more possibilities. Sutton (2006) further critiques modern technology and tools as being alienated from social and skill context, which turns the person using them into an operator.

2.4 Related Work

In this section, I will focus on related work within the domain of HFI and SCD. The projects in question are similar to this one in terms of methodology, critique and vision of the future.

2.4.1 Edible Speculations

The first project that I will examine is by Dolejsova (2018), and concerns SCD in the field of HFI with focus on participatory design. Edible speculations are a methodological approach for research within the field of HFI. The framework endeavours to include non-designers in the process, as a way to discuss the uncertain futures around food and base them in everyday food practices. What I found particularly interesting is the critique of SCD, and especially the Showroom approach to be detached from the public (Dolejsova, 2018). Dolejsova (2018) further stresses on the potential of SCD to challenge the current technological-solutionism stance in food practices. They invite non-designers into the process through workshops. One of them lets the participants prepare a sauce based on their life story, focusing on storytelling as seen in Figure 1.



Figure 1. From StreetSauce [image], by Dolejsova M., 2018, Edible Speculations: Designing for Human-Food Interaction, (https://www.researchgate.net/publication/330854615_EDIBLE_SPECULATIONS_DESIGNING_FOR_HUMAN-FOOD_INTERACTION)

This project was especially helpful in the initial stages of this thesis, with framing of the topic. The author highlights current gaps within the field of HFI, which allowed me to base my topic. I was also inspired by the process of critiquing current approaches and combining participatory design, in the shape of workshops.

2.4.2 The Urban Food Revolution

The Urban Food Revolution is a project carried out by Platform STAD. In this initiative, the focus was the future of Malmö and how it would look if it was self-sustaining. While this project is on a higher-scale, looking at the city, rather than a single household and kitchen, it provides insights on the issues of food security and community building. As the authors themselves recognise, cities are vulnerable and unprepared for the catastrophes that surround us, such as the fragile supply chain (STAD, 2024). As such, the project preaches for a self-sustaining city, as the benefits it brings extend beyond improving the environment, but also growing the community (STAD, 2024).



Figure 2. From The Heritage Grain Bakery [image], by Platform STAD, 2024, Malmö Urban Food Revolution, (https://storage.googleapis.com/matstaden/MalmoUrbanFoodRevolution.pdf)

One of the proposed ideas can be seen in Figure 2, called The Heritage Grain Bakery. The essence of the proposal is that in the future locally grown grains can be directly turned into local bread. As such, the process connects the farmers, the bakers and the consumers (STAD, 2024), closing the gap in the food sourcing process.

2.4.3 Human Hyena

The Human Hyena project is one which encompasses the concept of re-designing the human digestive system, to resemble the one of a hyena, which can consume any food. In this project, Gong (2014) is critiquing the problem of food waste, which is getting worse with each year. As such, the project asks

the question of how we can modify the human body in the future to adapt to the changes of the environment. The way the device works is by inhaling genetically modified hyena and human bacteria in order to change the digestive system. There is also a device which utilises enzymes that turn any smell sweet. All ideas explore tools as an extension of the body and expanding what we can do, as seen in Figure 3.



Figure 3. From The Human Hyena [image], by Gong, P., 2014, Synthetic Biology Project, (https://www.paulgong.co.uk/human-hyena)

I found this project through the cookbook by Cerpina (2022), and the concept of eating rotten food. What I found interesting is the critique that some cultures already consume fermented food, that may appear inedible because of their smell, such as old cheeses, kimchi, surströmming etc. Because of their peculiar smell we may stay away from these foods initially, as we have not encountered them in our own culture. As such, the critique and the Human Hyena project are interesting to think about our interaction with food depending on our background and how tools can extend our body's abilities.

3 Methods

In this chapter I will present the methods that I utilised to carry out the design process. I begin by examining Speculative and Critical Design (SCD) and the aspect that differentiates my approach, encompassed in co-design. Then, I differentiate three stages of the process, explored in the different subsections. This chapter also concerns the ethical considerations of the project.

3.1 Speculative and Critical Design

As implied in the previous paragraphs, this project focuses on the ambiguous futures and the role of technology. Normative design tackles the present, but is also pressured from industry and market expectation. For this reason, designers often turn to the approach of Speculative and Critical Design (SCD), fuelled by imagination of the future, and purposefully provoking new perspectives in order to solve what Dunne and Raby (2014) refer to as "wicked problems". It is through design based in everyday life, that we can cause profoundly disturbing effects on the people, to encourage reflection and propose alternatives (Dunne & Raby, 2014).

It is important to also distinguish the kinds of design in the term, and why they work well together. To begin with speculative design, it is mainly focused on the future, asking the questions of "What-If?". Dunne and Raby (2014) further preach the importance of speculative design as a critique of the integration of technology in our lives, especially highlighting the narrow view machines have of us - the humans. When it comes to critical design, it is focused on the status quo, primarily stressing the weakness in the current and offering alternatives (Dunne & Raby, 2014). It is in this part where we can implement a more prevalent societal issue, in order to generate thought in the viewer. Critical design is dual in its nature, for it explores values from a different world, but is materially located in the current world (Dunne & Raby, 2014).

Dunne and Raby (2014) highlight that in order to create a vision of the future, it is important to borrow techniques and tools from other disciplines beyond design. In this current project, I employ co-design approach, for its opportunity to involve non-designers in the building of the future.

3.1.1 Co-design

Designers have the power to create culture with their design (DiSalvo, 2021; Dunne & Raby, 2014). This is the power of design, which is also politics and should be handled well. However, with industrialization and the rationalisation of all practices, science became the most trusted, as implied before. Escobar (2018) highlights that societies were being operated solely by expert knowledge and science, essentially rationalising societal processes. Designers have recognised this tendency and have begun practicing participatory design, to include non-designers in the process, as a way to challenge the expert-driven process (Escobar, 2018). When building the future, it is important to consider what the actual people want and need, and not relying on science, this is where the gaps and catastrophes stem from. In this project, I utilise collaborative design methods and their potential to include non-designers in the process of critiquing the present and my future scenarios to create a vision.

3.2 Discover phase

The first phase of the design process is the discovery phase. This stage is characterised with extensive research of the field, in this case the kitchen, and its inhabitants. As such, it is important to bring forward the invisible that may be hiding in the routines of the home. After gathering the most important insights, I will use them to create scenarios of the future that will be used both for design inspiration and tool to talk with the participants about their view of the future.

3.2.1 Contextual inquiry

Contextual inquiry is the act of immersing into the place where an activity takes place, observing and asking questions to the participant (Hanington & Martin, 2019). This method is usually used to research workplaces; however, it is relevant in this study to understand the processes that happen in the kitchen, from routines to cooking. It can be used to discover the hidden processes in the activity (Hanington & Martin, 2019). This can then be used for design inspiration, or in SCD it can also be used to represent the hidden aspects of everyday life (Dunne & Raby, 2014). In this project I utilised contextual inquiry in order to enrich the speculative scenario that will be used to inspire the design.

3.2.2 Artifact analysis

To create a future scenario and speculative artifacts means considering a great deal of detail about the society and its values. In order to offer alternatives to current technology it is important to understand it and its impact. Artifact analysis is a method that assists the understanding of an object's context within society (Hanington & Martin, 2019). The question that drives this analysis is "what do objects have to say about people and their culture, time, and place?" (Hanington & Martin, 2019, p. 14). This method is a great fit for this project not only to understand what current and past technology have to say about society, but also provide a framework in the construction of prototypes that reflect the future scenario.

3.2.3 Scenario

To design alternatives is crucial for SCD. The design is also based in an imagined future. As such, we create a scenario for it. Scenarios as a method are used to imagine how a product will be used in the future and are usually written from a specific point of view of a persona (Hanington & Martin, 2019). Nevertheless, in SCD, as Dunne and Raby (2014) describe it, scenarios, and more specifically What-If scenarios, can be used as thought experiments to imagine how society would react to catastrophes or certain circumstances. Per se, scenarios are fictional for SCD, which as the authors critique, is seen as a stigma within the design community, because design is so absorbed in the industry (Dunne & Raby, 2014). However, it is important to recognise the

need to imagine and dream about the future, not only to bring innovation but also to critique the current state of affairs in technology design, which is the focus in this project.

3.3 Data analysis

The next phase in the design process is the analysis of the gathered data. The analysis assists me with locating important insights and opportunities.

3.3.1 Thematic analysis

Thematic analysis is a method of analysing fieldwork data and systematizing it in order to find patterns and themes across the data (Braun & Clarke, 2006). What is important with thematic analysis, as Braun and Clarke (2006) highlight, is to stay transparent about our assumptions and questions that guide the analysis. There are also multiple ways of conducting this analysis, ranging from straightforward and overall analysis of the data, to more detailed and interpretational ways of certain aspects. I utilise this method throughout the whole design process, from fieldwork to usability testing, to highlight the most important insights. The approach I take in this project falls in the more straightforward analysis.

3.4 Generation phase

The last phase of the design process consists of the generation of design and evaluating it. For this phase I utilise collaborative brainstorming, prototyping and sketching, which allow me to quickly imagine and discard ideas, while choosing the most notable ones.

3.4.1 Collaborative brainstorming

Brainstorming as a method for generation allows participants to generate ideas in relation to a prompt, with the aim of generating as many ideas as possible (Wilson, 2013). I use collaborative brainstorming for this project, in order to generate ideas quickly and invite as many perspectives as possibles. The prompts for the sessions are the scenarios. I use materials ranging from paper to clay, in order to encourage creation of 3D ideas. One of the most important things is to facilitate the session well and encourage people to create (Wilson, 2013). Encouraging play with clay allowed the participants to explore more ideas and escape the constraints of what's possible.

3.4.2 Prototyping

The power of SCD lies within its ability to disturb the viewers, because of its situatedness in everyday life (Dunne & Raby, 2014). Thus, it is important to manifest the future into a design that the viewers can interact with. Through prototyping I plan to transform the idea of this alternative future into something material that can be interacted with. As such, this kind of prototyping can be characterised as prototyping as the means of inquiry, as it

seeks to create knowledge (Matthews & Wensveen, 2015). Matthews and Wensveen (2015) further compare this kind of prototyping to the notion of "provotyping", or prototypes that provoke reaction from the users. This is especially relevant for this project, as the point of the future artifacts is to encourage reflection in the viewer. When it comes to the difference between provotypes and critical design artifacts, as Boer and Donovan (2012) highlight, is the active engagement in understanding the field, in order to critique the interests in the design process. Similarly, Dolejsova (2018) has critiqued the authoritarian practice in SCD, which has been separated from the field. In this project, prototyping is used to inquire not only the response of the participants about the future but also create knowledge about the field, drawing upon the past and the speculative future.

3.4.3 In-situ testing

Lastly, in order to generate insights and response with the prototype it is important to deploy it in the place where it is supposed to reside. As this project is focused on the cooking practices it is focused on the kitchen as well. As such, prototypes will be tested in-situ in the kitchen when possible. What is important to note for speculative artifacts is that it is important to invite the participant to imagine the alternative future and not just be a passive recipient of the artifact (Dunne & Raby, 2014). As such, the artifacts need to be exaggerated in their form and appearance, to convey their strangeness to the world they are situated in (Dunne & Raby, 2014). At the same time, it is important to consider the implications of testing speculative artifacts. As Fordyce (2021) highlights in their paper, testing speculative artifacts is testing the legibility of the message within them. As such, it is important to consider how I invite the people to interact with the

3.5 Ethical concerns

The research in this thesis followed the ethical guidelines provided by the Swedish Research Council (Stafström, 2017). As such, I made sure that the participants were well informed about the purpose of the study and their participation, through the informed consent form (Appendix 1). What was important was to make sure the participants felt safe during the research, and also have the opportunity to step back from the study.

The participants remain anonymous in the study and their personal data is stored securely where no third party can access it, complying with the General Data Protection Regulation. The participants are regarded as "Participant N" to remain anonymous and be easily differentiated. Furthermore, any audio recordings or pictures were taken with the participants' permission and have been altered to remain anonymous.

4 Design process

The core of this thesis is the design process. It is characterized by four phases. Discovery phase, where I explore the field now and understand past and present practices. Generation phase I, where I speculate about the future through the insights from the fieldwork in the shape of scenarios. Generation phase II, where I reflect on the insights and narrow down my ideas, through thematic analysis. Prototyping phase, where I transform the insights into artifacts which I use to talk to the participants to evaluate how provocative and critical they are. The process is illustrated in Figure 4.

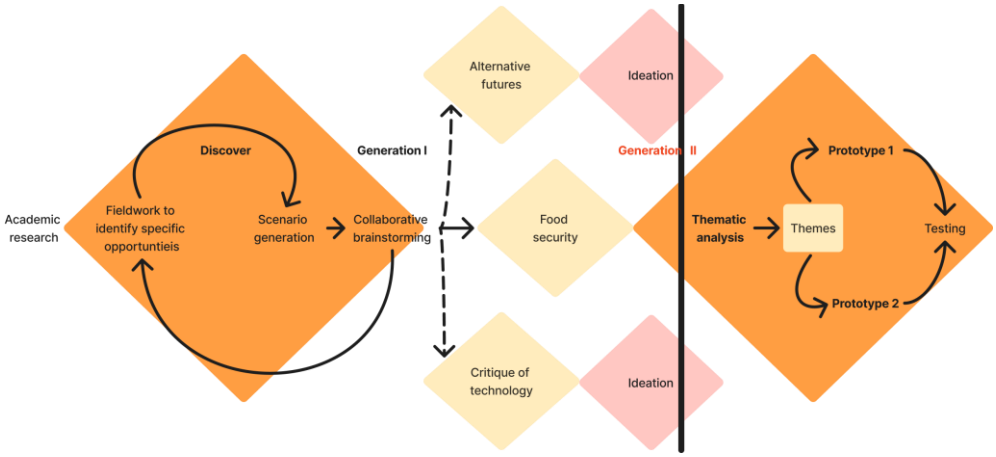


Figure 4. Illustration of the design process

4.1 Discover phase

To begin the design process, I needed to get out in the field and meet the people and their practices. The purpose of going into the field is to understand how things are done now and how design supports and facilitates certain practices. The discovery phase is structured in the following way. First, I observed and participated in cooking practices. Next, I analysed the stove as an artifact in the past and the present. Then I created scenarios with the insights that I gathered which I used to talk to participants about the speculative future.

4.1.1 Contextual inquiry

To begin the design process, I initiated a field study of cooking. This included two contextual inquiry explorations, where I also participated in the cooking process, one more actively and the other more passively. Through this study I wanted to pinpoint specific opportunities within current cooking practices.

Research Setting

I participated in two cooking practices and each one took around an hour. One was video recorded, with the participants permission, and for the other one I took a few pictures. The participants are from different age groups and

are both immigrants in Sweden. During my observation and participation, I was asking questions and asked the participants to speak their mind when possible.

Outcome

In the first study, I took an active role in participating in the making of Dubai chocolate cake jars. The participant created the recipe themselves, by observing the trend and combining recipes together. The participant was also more experienced in baking compared to the other participant. The process can be summarised in Figure 5. The process of making Dubai chocolate cake jars summarised

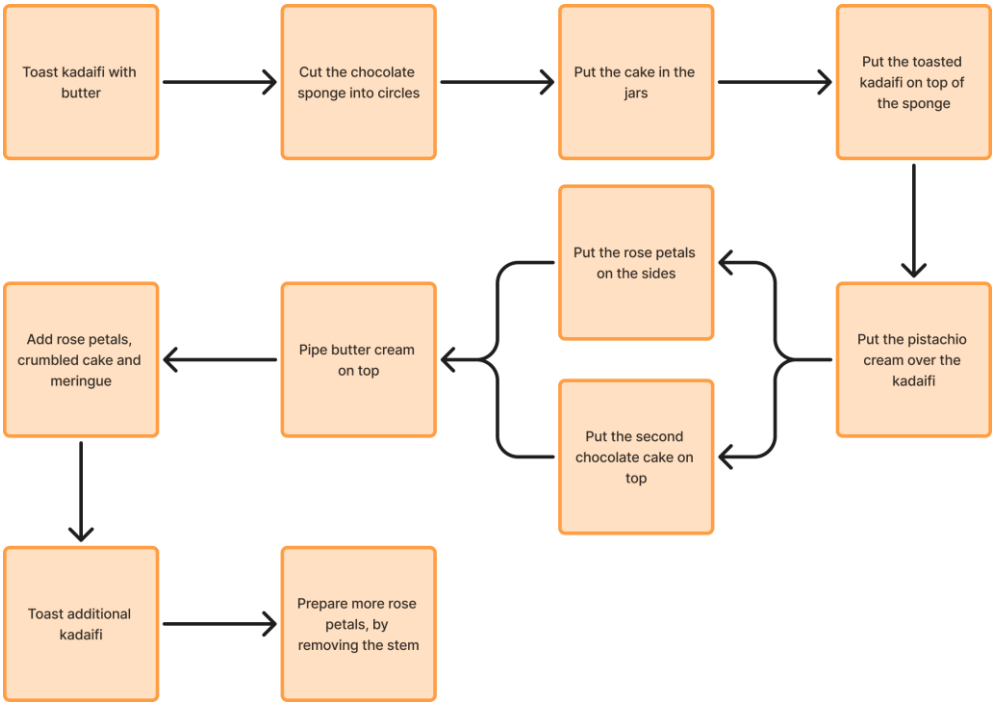


Figure 5. The process of making Dubai chocolate cake jars summarised

An important insight from this study was that the participant discovered the ingredient of kadaifi for the first time. They expressed curiosity and interest about it. As we needed to toast it, the participant did not use precise measurements, despite working with this ingredient for the first time.

Another interesting insight was that the participant used a spoon that they altered, specifically for assembling cake jars. While they could not locate the original one, the participant grabbed a spoon and bend it, as seen in Figure 6. It was interesting to see how the participant adapted tools to their needs, coping with the resistance in a way.



Figure 6. The bent spoon

In the second study, I took a less engaging role, by suggesting alternatives and giving hints from time to time. The study was based in my kitchen, in order to determine how the cooking experience changes in a different kitchen. The participant was making tofu stew and the process was summarised in Figure 7.

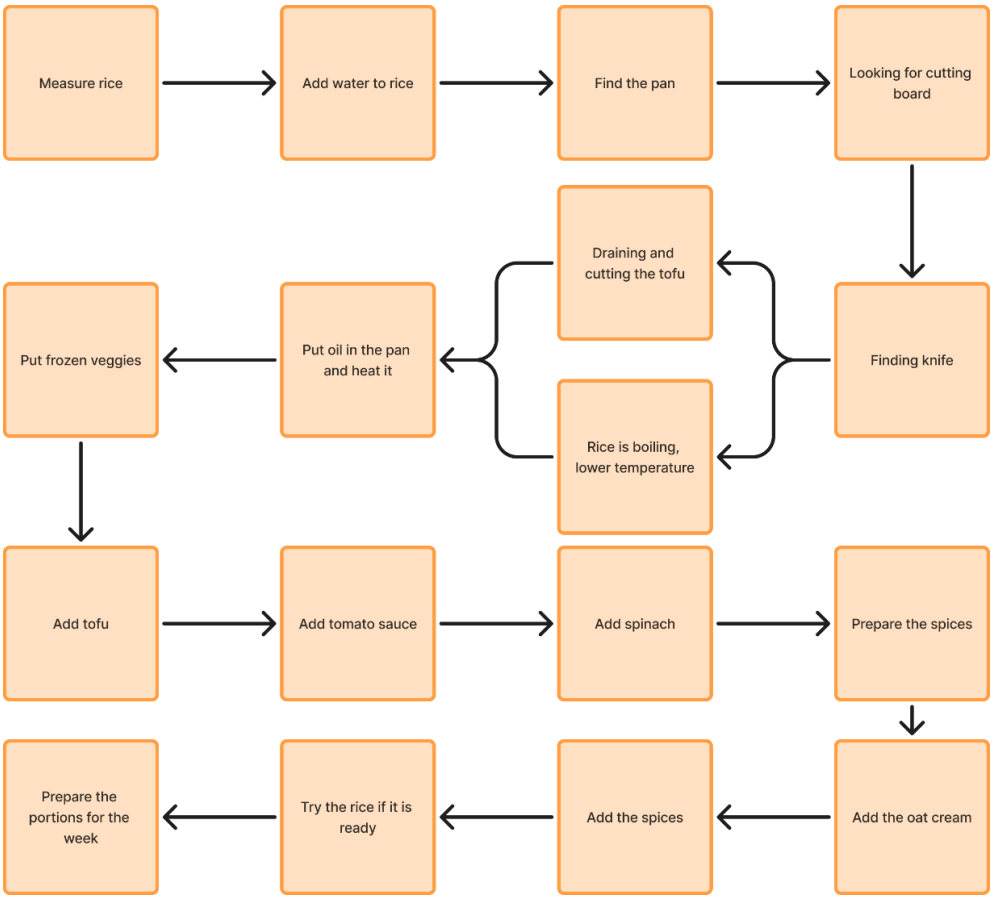


Figure 7. The process of making tofu stew summarised

The first important insight from this study is about locating the tools and ingredients for the process. More specifically, the way the participant located the cutting board was by its shape. They explained that the shape of the cutting board led them to a long narrow cabinet, as seen in Figure 8. In their paper, Sutton (2006) speaks about recipes as memory-jogs during cooking, and it is interesting to compare it to this instance, where the tools themselves act as a reminder as to where they could be.



Figure 8. Explaining the shape of the board hinting towards where it stays

The second interesting insight is in relation to the measuring in the recipe. The participant expressed a few times how they like to follow instructions and measurements precisely, in order to get the same result every time. This applied for their preparation of the rice and the portion sizing. However, there were instances where the participant was adding things by eyeballing, as seen in Figure 9. Such ingredients were the green onion, the spices and the oat cream. In such instances, the hands were the measuring tools, either by grabbing a handful or by pouring and feeling the difference in the oat cream container.



Figure 9. Difference between measuring onion by handful and rice with a glass

4.1.2 Artifact analysis

Creating a scenario for the future means understanding how the values in the society will change and how the process evolves. My approach in this project includes looking back in the past and comparing it to the present, highlighting the changes and values that design has brought, highlighting its power. This analysis was inspired by the study of the Swedish kitchen by Hemmens forskningsinstitut (Hemmen's Research Institute), conducted in the late 1940's. They have published a short video about the study, which is in Swedish, so understanding it was a bit difficult. Nevertheless, something that was interesting was the oven that was used back then. They showed a

Husqvarna wood stove model from 1926, which can be seen in Figure 10. It looks quite different compared to a conventional modern oven today, Figure 11. As such, I wanted to analyse their difference and draw a conclusion as to how the new design embraces new values for the society and its effect on family dynamics.



Figure 10. Husqvarna Vedspis 1926, adapted from Ett praktiskt kök [Video], by Hemmets forskningsinstitut, 1953, YouTube. (https://www.youtube.com/watch?v=7v2ZxNgaRis)



Figure 11. A conventional electric stove and oven in my apartment

To begin with, I started analysing the wood stove. It is important to note that my analysis was based on images and videos, because of the inability to find such a stove today, which is also important for the analysis. In addition to

analysing this stove model I will also focus on analysing a wood stove that I have seen in real life, Figure 12. The first thing that can be noted is that they look very similar. Both stoves have a fireplace, ovens, stoves and exhaust that pumps the smoke out of the house. Both used wood as fuel, which also meant slow cooking, which is preferred amongst many people. The detailed analysis can be found in Figure 13.



Figure 12. Old Bulgarian wood stove

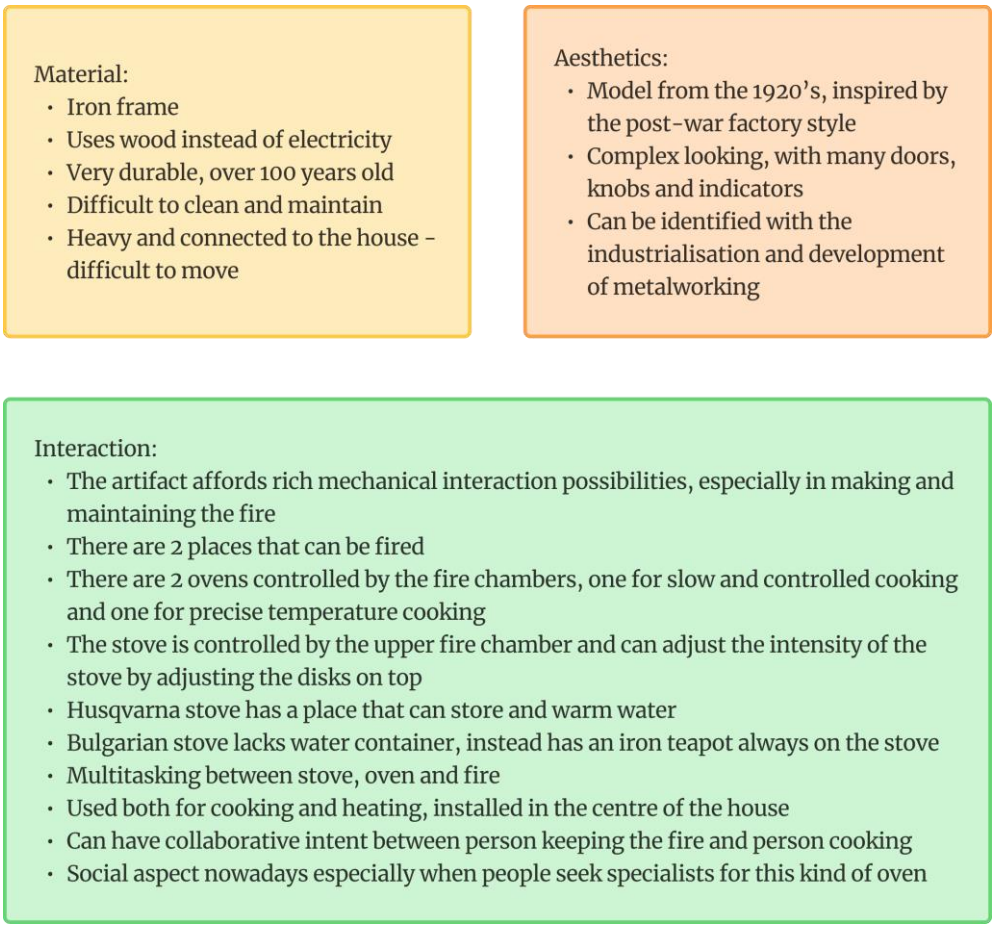


Figure 13. Artifact analysis of the wood stove

I used the guideline from Hanington and Martin (2019), focusing on materials, aesthetics and interaction to analyse the artifact. It is also very important to consider the location of the artifact on top of these aspects. To begin with, these ovens were heavy and hard to install and move. They were used not only for cooking but also heating and as such were located in the centre of the house, which has been the kitchen. The kitchen is private from the outside world, but is the social hub of the home, where all activities are done. Nowadays, it can be found in use in the north of Sweden and in some villages in Bulgaria where electricity is also scarce.

This old design encompasses the values of the society back then. First of all, the oven was used to heat the house, which meant people were gathering around it, it was a social hub. Then, maintaining the fire meant being skilful in working with it and being patient, as heating the stove was a slow process. Slow cooking also suggests that the pace of life was slower.

Looking into this old stove also made me curious about other types of cooking methods. I present a few of them without a detailed analysis in Figure 14.

	Name	Time	Characteristics
	Stone oven	• Ancient times until the iron oven • Now used for artisanal pizza/bread	• Heats slowly • Interesting flavour because of the wood • Used for larger meals
	Pirostia	• In the past, until the iron oven • Today almost no one uses it	• Used directly over the fire • Superstitions around it • Easily transported
	Geothermal cooking	• In the past • Today is still practiced in Iceland and other regions in the world	• Slow cooking • Superstitions around it • Utilises natural resources
	Swedish torch	• Used in the past during wars • Nowadays is a survival technique	• Efficient use of wood • Easily crafted • Fire off the ground

Figure 14. Other cooking methods and short analysis. Geothermal Cooking image, from Lava Baking And Geothermal Pools At Fontana, Iceland [image], by Hand Luggage Only, 2025. (https://handluggageonly.co.uk/the-1st-day-in-the-land-of-fire-and-ice-iceland-lava-baking-geo-thermal-pools-and-the-golden-circle-part-1/). Swedish Torch image, from How to Make a Swedish Fire Log [image], by Fresh Off The Grid, 2020. (https://www.freshoffthegrid.com/swedish-fire-log/)

All of the above methods for cooking have caved the way to the modern electric stove. What is common among all of these is the need to start with fire, with the exception of the geothermal cooking which relies on the temperature of the ground.

The modern design of the stove offers an efficient and quick way of cooking. However, it is stripped off many of the values that people 100 years ago held. In a way, it is impoverished in terms of experience and interaction. The values that today's stove encompasses are efficiency, swiftness, and precision. While it is efficient and easy to maintain, it is important to consider what we sacrifice with this design.

4.1.3 Scenario

After my exploration of the stove design of the past, I gathered a few insights as to what it affords and the values of the people that used them. The insights together with my previous research on other cooking methods allowed me to characterise the past, through the wood stove. I identified the characteristics of the stove and the main value that it encompasses. In the case of the wood stove, it is self-sufficiency, because people were taught how to gather and prepare things from scratch. The detailed analysis can be seen in Figure 15, where I also analyse the present electric stove and the imaginative future stove, which I use to materialise the values of future societies in my scenario.

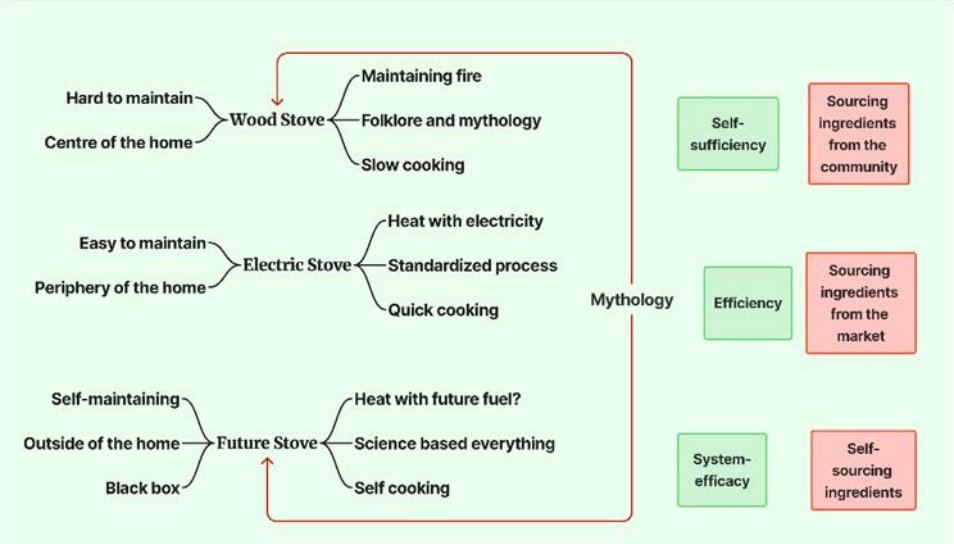


Figure 15. The evolution of the stove and the values that the design encompasses

When I analysed the modern electric stove there are 2 important insights. First of all, the value of the design is efficiency, characterised by the quick cooking process and ease of maintenance. Efficiency was the leading factor for design, as an endeavour to free the housewife, so she can have time for herself (Bell & Kaye, 2002). Second, science has become the leading force in people's lives, from health care to cooking. As a result, people and technology have become more rational. This has limited the role that folklore and myths have on people's everyday life. Traditions that some families keep preserve these beliefs. For example, some people in Iceland still prefer to bake bread in the ground.

Now, when I imagine the future, I follow this pattern of progress. The question that led my exploration is what goes beyond efficiency (Appendix 2). If the purpose of efficiency was to free the housewife from the arduous domestic work, then we can imagine a future where household members, not only wives, are freed from any kind of domestic labour. The kitchen is separated from the house, as its own factory that can sustain itself. In this future, the value of the design is self-efficacy of the system – being confident in one's abilities to produce intended results (Bandura, 1977). The kitchen in itself, will become a self-sustaining ecology of machines that we can just control by pushing buttons. In other words, it will become a black box, as all the technology inside will be made to act as one to produce the final product (Latour, 1987) without human intervention. To elaborate on the black box as a metaphor, it is possible only for experts to maintain and explain how it works. As such, I envision that as people drift further and further from the kitchen and their understanding of it, folklore beliefs and myths will rise again to explain it, as illustrated in Figure 15.

On the other hand, I also imagine a future, where we have experienced a crash of the supply chain, hence being forced to go back to basics, such as food

sourcing and cooking on open fire (Appendix 3). With such scenario I want to explore ways of bridging the gap between people and their food, as well as address the issue of food security. The scenario is based on a society of the future that is so isolated from its food that they have to “re-discover” some basics, or be supported in doing so. It also suggests a future where collectivism and collaboration are central and important aspects of sustaining and co-creation, as well as delegating tasks.

4.2 Generation phase I

After studying the field and gathering insights regarding cooking it was time to brainstorm design ideas related to the imagined futures. In doing so, I incorporated co-design principles and facilitated collaborative brainstorming workshops where I let the participants reflect on the scenarios and the given prompts, in order to imagine future designs.

Research setting

I invited 2 participants and each workshop took 1 hour. It was audio recorded, with their permission, and I took a few pictures. The participants are different age groups and are international students. Figure 16 shows the materials I provided for the session. I provided the participants with modelling clay, as a way to stimulate their 3D idea generation, pens and paper, and sticky notes.



Figure 16. Materials used in the workshop

The workshop began with me providing the participants with the prompt questions and reading the scenario out loud. Then we had several discussions regarding different topics, ranging from the scenario itself to the brainstorming prompts.

4.2.1 Collaborative brainstorming #1

As part of my transition from present to future ideation I conducted a collaborative brainstorming session. Through this exercise I wanted to test the comprehension of my current scenarios and see if they can stimulate the participants to think about their coping skills for the future.

Outcome

The discussions and the creations inspired very interesting insights. The first insight that is important to note is the mistrust in collectivism in times of crisis. In the scenario, the main character was collaborating with his neighbours, which the participants found weird and mentioned they would not do it. One of the participants even referenced a study where tsunami survivors were collaborating initially, but eventually drifted apart.

The second important insight, is the reflection of Participant 1 on the current focus of companies. Participant 1 mentioned that now companies focus beyond efficiency on revenue:

> "...we are going beyond efficiency to revenue, for example adding AI to anything, not because it has purpose for the user but because of revenue"

The participant acknowledged that this might be out of the scope of the current scenario, falling into my other future idea.

Lastly, the participants were thinking about alternatives and ways of turning materials and technology they have access to into something else. For example, using bikes in order to generate mechanical power, or turning their ovens into fire pits. As seen in Figure 17, both participants had different ideas of a "fire pit", when asked how they will cook without electricity.

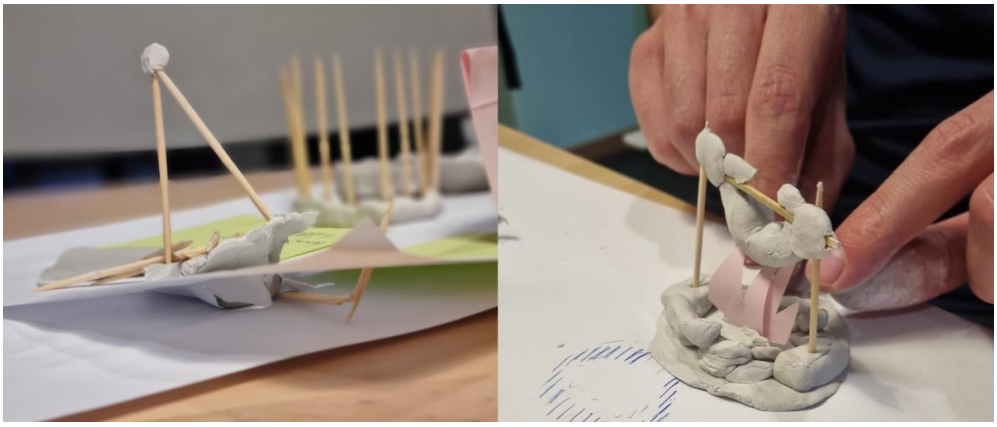


Figure 17. Different concepts of fire pit

4.2.2 Collaborative brainstorming #2

I repeated this method and organized a short brainstorming session revolving around my second scenario. The aim of this exercise was to check the clarity of the scenario and the reflections that the participants had.

Outcome

The participants had difficulties bringing ideas to life from clay and even drawings, as seen in Figure 18. Nevertheless, there are a few very interesting insights, from the discussion. The first important insight is from Participant 2, who expressed their belief in history going in circles, regarding the state of the future kitchen. Their reflection was that in the past the kitchen was the centre of the house, especially for people in lower class, but then it moved to being in a separate wing, especially in royal houses. They gave an example of the Versailles where the food would arrive cold, because of the distance that had to be covered by the servants. Then as we moved to apartments where the kitchen was again within the boundaries, because we changed our way of living. Participant 2 further reflected:

> "History is going in cycles. Kitchen is moving out again, but instead of people who were poor and worked for you, now you have a machine."

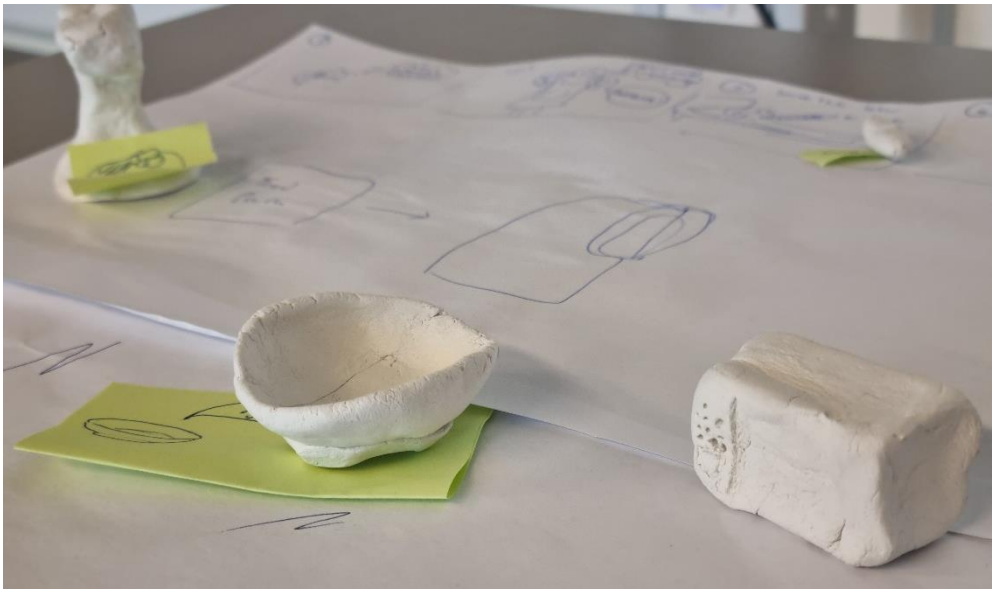


Figure 18. The ideas that the participants came up with in workshop #2

The second insight from the discussion was that the scenario was seemingly contradicting for them. As such, Participant 3 asked why should we bother cooking if we can just order food. Then they tried to make sense of the reason to have a kitchen, relating it to issues of logistics. They reached the conclusion that the future kitchen will know what nutrients we need and choose the

recipe, regardless of our intention in the kitchen. Participant 3 then commented on the idea of cooking in the techno-centric kitchen:

> *"There would still be an attempt from technology companies to give the idea that there is some kind of ritual (as cooking). Preserving the humanness, but in a very shallow way, just to keep this going forward."*

The third important insight that the discussion produced was regarding the vision of Participant 2 regarding the future. The participant expressed her belief in a future, where machines will not be evil and ruling, but will instead take the role of a nanny, showing us how to do things, hinting towards a collaborative future.

4.3 Generation phase II – Reflections and way forward

The field study provided a lot of insights into the context of cooking and the reflections of the future. Through thematic analysis I located 3 important themes, within the topic of the future of the kitchen and cooking: Technology as its own agent; Adjusting the cooking tools to meet needs; and Feeling through the body. These themes are interconnected with other smaller themes, which are also worth mentioning, as seen in Figure 19. As mentioned before, it is important to characterise the process of thematic analysis for this specific project and be transparent about the questions asked. The thematic analysis I conducted could be characterized in between the two polars of the method. On the one hand, I used the constructionist method to make sense of how society, or more specifically the changes of the design, affect the experiences and the meanings people make (Braun & Clarke, 2006). On the other hand, I also take an essentialist stance in finding patterns and themes within the insights, characterizing experiences in a straightforward way (Braun & Clarke, 2006). It is also important to mention that the gathering of the data and the analysis were led by the main RQ and the mission of SCD to cause reflection on our current technology mediated present. In the following section, I take each theme and try to build a prototype resonating with it.



Figure 19. The three themes extracted from the fieldwork

4.4 Prototype 1: Preservation Scale

4.4.1 Prototyping

The first prototype I explored was in relation to the apocalyptic scenario, where people have to source their food and re-learn the basics. An essential part of being sustainable is being able to preserve food. With this in mind I wanted to explore a way to assist people in their exploration of preservation methods and food's longevity, and also encourage people to reflect on their past experiences. Past experiences here refer to any tacit knowledge passed down in their community or family. My inspiration was of an old artifact that is almost impossible to find in modern households: a brass scale. Nowadays we rely so much on the digital scale that we have found ourselves deprived of the ability to estimate ingredients and are located in a standardized trap. The brass scale can be used to compare the weight of things, so I wondered what it would be like if instead of weight it reflected the longevity of food and how certain ingredients affect it. Figure 20 shows the sketch of the prototype and Figure 21 shows the final look of the prototype. The prototype was built quickly and is low-fidelity. It is built using cardboard and is controlled by an Arduino and a servo motor. Because of its fragility I used paper with different food graphics, instead of real products.

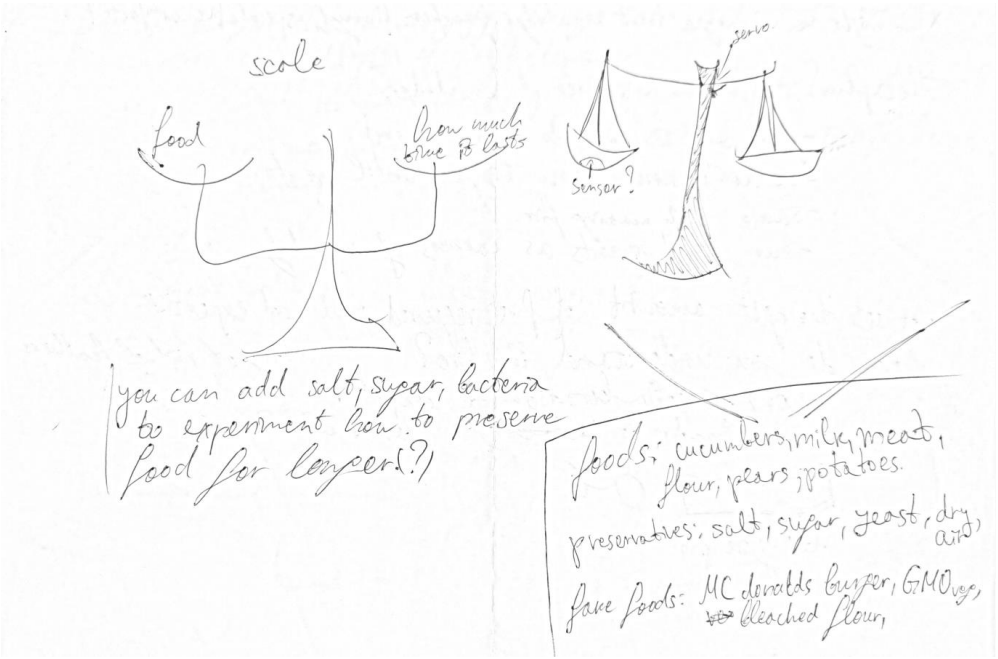


Figure 20. Sketch of the preservation scale



Figure 21. The preservation scale's final shape

4.4.2 In-situ testing

Research Setting

I tested the Preservation Scale 3 times and each one utilised a different testing technique. Testing 1 and 2 were done with 1 person each and the scale was controlled by me and my assumptions on food, with the difference of how I presented the foods to explore, as seen in Figure 22. Testing number 3 was done with 2 people, who alternated their control of the scale. The participants are from various international backgrounds. The sessions were audio recorded with the permission of the participants. The testing took around 20-30 minutes and it followed a similar structure: inquire about their knowledge on food's longevity and preservation methods, then let them explore

combinations and possibilities. I also revealed that the scale is speculative and let them think about it in an abstract way.

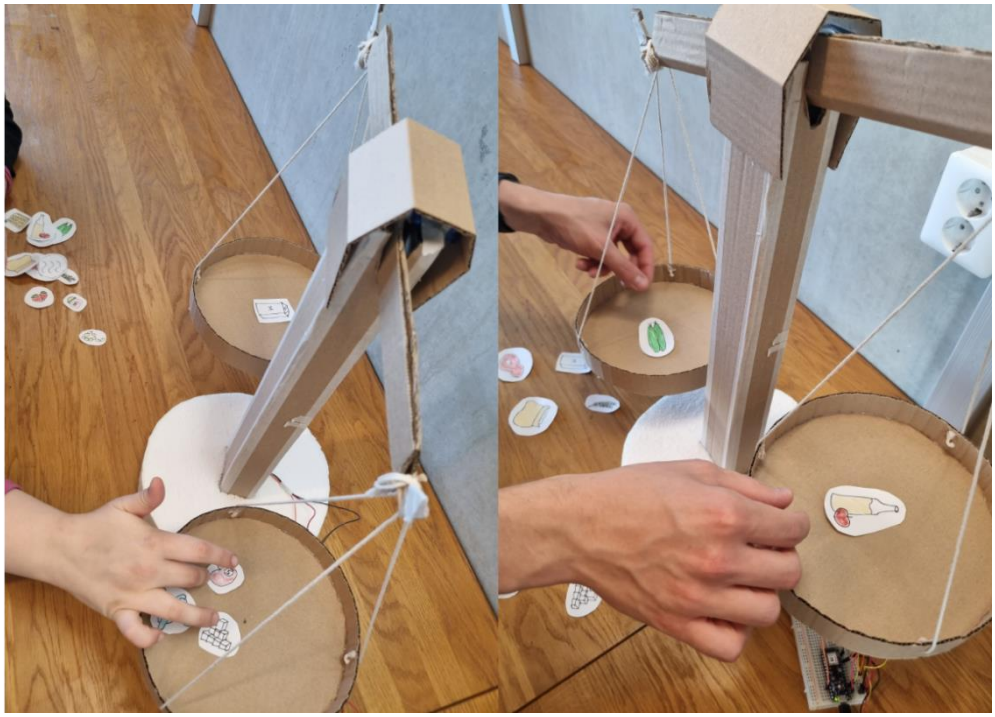


Figure 22. Testing the scale. Left side, comparing foods. Right side, comparing longevity with preservation methods

Outcome

There are several interesting insights that came out of this testing, mostly in relation to the participants' reflection on their knowledge. As such, one of the most important insights is that they have seen their families preserve food in different ways, but they themselves have not done it. For example, Participant 4 reflected how their mother would preserve milk in the shape of labneh balls submerged into oil, expanding the preservation methods around milk. Similarly, in testing 1 and 2, the participants came from similar backgrounds and they both reflected how their grandparents would preserve tomatoes as a tomato sauce sealed in jars. As such, the discussion around their experiences brought traditions and tacit knowledge in light. Participant 4 further reflected on foods they have seen:

> There is one thing that I'm not sure if it's something that they make to preserve food, but it looks like flour… And you put this in a pot of water and it becomes soup consistency, and it is delicious.

The second important insight is that many of the participants did not understand how preservation even worked. Participant 5 even reflected:

> "How do you find bacteria in the wild?"

As such, the prototype allowed them to test their assumptions and bring them closer to the process in a way, but it did not provide full understanding or bridge the gap as intended. Furthermore, an interesting reflection from Participant 5 was the fact that when they were looking for recipes for jam, which has been done for centuries, they required extra products and materials, complicating the process and making it overly standardized:

> "Yeah, then you can just buy this and this and that will like preserve it. But I'm like but I want to make it at home"

Lastly, in the first testing, Participant 6 reflected that using scale did not make sense to them, if it is about comparing food and preservation methods. They reflected that I do not even need to use food, just food options:

> "…food option number 1: local, non-GMO, freshly harvested, the cost. Information that people care about. And then you put maybe less ethical version. And then you try to balance it out."

This insight suggests for iteration of the prototype with a new goal of not comparing, but trying to balance the scale using different parameters.

Insights

The final evaluation of this prototype is that while it helped the participants reflect on their food practices and knowledge in terms of preservation, it did not fully connect them. As such some participants were even asking how certain methods work or are even sourced. Nevertheless, as a discussion tool, the prototype allowed the participants to re-discover traditions and knowledge passed down in their family/community, as seen in Figure 23.



Figure 23. The two participants discussing food practices

Furthermore, the reflection of Participant 6 about the usage of food options as opposed to actual food is an interesting direction for iteration. By giving food options with characteristics that oppose each other, the participants could try to relate to actual foods that they buy and mirror these options. Then, trying to balance it out with things, such as cooking, sourcing of preservation methods can bring people to think about even more examples from their everyday life and perhaps bring them closer to this knowledge.

4.5 Prototype 2: Toastless Toaster

4.5.1 Prototyping

The final prototype for this project is the Toastless Toaster, ideated within the theme of technology being its own agent and technology handling all food. The idea is further inspired by the critique from Shapiro (1986) as cited by Sutton (2006, p. 97) of bread machines and their ability to produce bread "which no human hand has touched from the time the wheat was planted until it was taken from the pan in which the loaf was baked". Relating this to the issue of food security and the gap between people and their food, I sketched the idea of the Toastless Toaster, Figure 24. The idea is that the toaster is a black box that turns seeds into toast, handling all the processes in between, embodying the disconnection between people and their food. The input movements are inspired by gestures characterizing the process of milling flower and kneading dough and they are put there to create an "illusion" of cooking ritual, as per the insight from the second workshop. The toaster omits a light in relation to the input of the user, which is there in order to provide a feeling of control over the box. I chose light because it is easy to control and set up, compared to other feedback options.

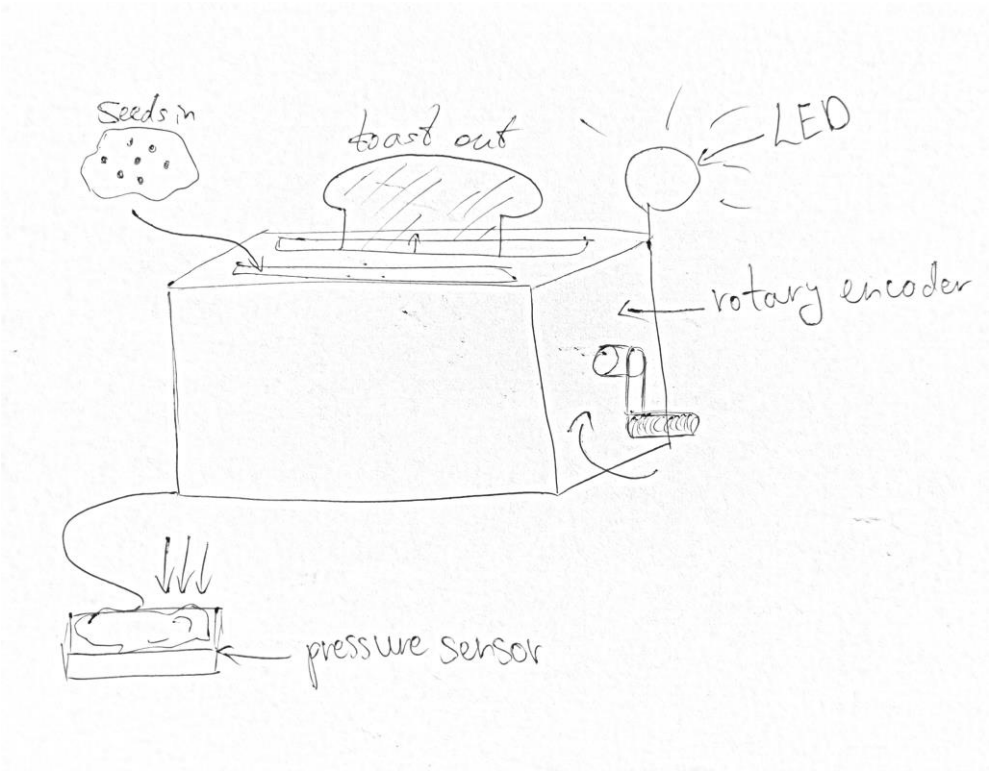


Figure 24. Sketch for the toastless toaster

The prototype is constructed out of laser-cut wood and cardboard and the final version can be seen in Figure 25. It utilizes Arduino Uno to control the input and the output. The input is measured through a rotary encoder and a FSR pressure sensor. The output is both the toast and light from a LED ring. The code that controls the input and the output has been generated by ChatGPT 4.0 through multiple prompts, for example:

> "Make the pressure determine the colour, or even the saturation, while the rotary encoder is responsible for the flickering of the LED. Make the servo motor rotate 60 degrees until the rotary encoder and the FSR get any input again, but with delay of 20 seconds, so the person has time to react and doesn't continue putting input".

However, often times the code was not acting the way I was imagining it, hence I was also adapting the code or even exchanging it with already written examples in the Arduino IDE Libraries.



Figure 25. The final version of the toastless toaster

4.5.2 In-situ testing

Research setting

The testing of this prototype was conducted in 4 sessions, each taking around 20 minutes, which were audio recorded with the permission of the participants. The participants are in their 20's and are design students. The nature of the prototype made the testing challenging, in terms of presentation. As highlighted by Fordyce (2021) testing speculative artifacts is inspecting the legibility of the message within the design. This could be done by disclosing the nature of the prototype, as speculative, or letting the user interact with it considering it with usability disposition. I explored both options for testing with different participants.

Outcome

First, I will elaborate on the testing sessions done without revealing the nature of the prototype, as speculative. The testing began by me giving them the instructions of how to prepare a toast, by feeding the toaster seeds. The participants put the seeds in the container, as seen in Figure 26. Then I gave them explanations on the types of input and let them play with the toaster for some time, as seen in Figure 27.



Figure 26. Putting the seeds into the toaster



Figure 27. Playing with the toaster

This group of testers provided interesting insights, in terms of how it felt to work with the material. One participant noted that the kneading motion felt very natural, compared to the cranking one.

When it comes to the perception of the message of the design, the participants grasped the main idea in a nuanced way. Participant 7 perceived the message to be that even if machines are automated, they will still need the human factor:

> “I don't think we can be replaced, even in making bread pretty soon. Because you still need at least to taste it, to see if it's good.”

The idea of tasting the final product is an interesting idea of evaluating things, which machines cannot do. When it comes to taste, it is interesting to explore how olfactory feedback could fit into the setting, as it is connected to the sense of taste.

Participant 8 saw the toaster as a tool that essentially bridges the gap between people and their food, especially in terms of making bread. They reflected that it is interesting, as nowadays we have the already prepared and cut bread, so going through the steps beforehand made them reflect on that lost connection. At the same time the participants also reflected that they felt control in the beginning by doing something that got the thing started.

Furthermore, Participant 8 mentioned how the appearance of the thing as a black box and its mystery, made them curious as to what it does. They also mentioned how it made them feel like they could experiment, a reflection over standardization in a way:

> You are kind of more experimental with it, rather than just buy the same bread with the same ingredients and the same ratio of ingredients – this is going to be different. It makes me feel like depending on how I interact with it, it will come out different and depending on how many seeds I put.

The second group of participants were the ones to whom I revealed the nature of the prototype and invited them to think more abstractly and imagine. As such, they were really thinking about the ways in which the prototype makes them feel emotionally. Participant 9 reflected about the feeling of control over technology:

> “People get a satisfaction from mastery of the machine. Like I want to feel smart by using something I don't understand, but it listens to me.”

Essentially, we discussed the way people crave control and how black boxes can stroke the ego of the user. Participant 9 also compared the experience to the infamous Juicero design:

> Like that's an interesting idea of when you're actually looking towards this over-engineering as a way of not necessarily like giving people the experience of an authentic thing, but giving people sort of what is seen as more of an exclusive experience of you have to have this to ever be able to get that juice.

On the other hand, the last testing included two participants 10 and 11, Figure 28. Their reflection on the interaction was that they are working together with the thing to create something, similar to the first testing. Their reflection over the light was also different than the rest of the participants. For them the light represented the process inside the toaster and that there is correlation to things, but did not feel in control. Like the rest of the participants, they also mentioned that the message is about addressing our disconnection between getting something from the store and actually making it.

Insights

After testing and evaluating the clarity of the message within the toastless toaster, it is suffice to say that the participants were on the right track when it comes to understanding the concept. The major difference between the two testing groups is the feeling of control. In the non-disclosed group, the people felt in control of the machine, as opposed to the disclosed group. This difference could be stemming from the fact that the non-disclosed group was using the toaster with the mindset of an actual product, while the disclosed group was looking for meaning in the feedback and interactions of the toaster, hence unconsciously noticing the lack of connection to things.



Figure 28. Interacting with the prototype in a social setting

What is interesting is that most participants understood the basis of the message - the disconnection between us and our food, and that the toaster is an endeavour to connect us to the processes. Most of them appreciated this "new way" of interacting with a toaster. Nevertheless, for others it felt weird, as there was no order of interactions, which was partly the aim for this prototype. Speculative artifacts ought to be absurd and exaggerated in their "unreality", in order to invite participants to imagine (Dunne & Raby, 2014)

5 Results

How might we challenge the trend of efficiency in human-food interaction design, through speculative and critical design?

I challenged the trend of efficiency by creating two future what-if scenarios, with focuses on the isolation between people and their food, and the overall automation in the kitchen. I materialise these futures in 2 artifacts, that reflect the future values, in an attempt to cause reflection.

1. How might we prepare people for the ambiguous future surrounding food practices through SCD?

While underdeveloped, this RQ is addressed through the Preservation Scale concept, as an endeavour to bring people closer to the tacit knowledge that is passed down in their community and also allow them to explore preservation methods. With preparation, it is not only about preparing physically, but also consider the possibility and reflect on one's readiness. Such reflection was

achieved during the brainstorming sessions with the scenarios as well. In this sense, a well-written and detailed story can make people feel and imagine as if they are in it, acting as provotypes in a way.

2. How might we critique the current mediation of technology in the kitchen through SCD?

I address this RQ with the Toastless Toaster and the metaphor of the black-box as its own agent. While it is not a new metaphor, exaggerating the inorganic nature of the black box in the context of food has the potential to disturb people. Most of the participants reflected about the disconnection between them and the process of making bread, in the context not only of technology but also of supermarkets' abundance and standardized nature.

6 Discussion

In this section I will discuss the outcome of this study. First, I will highlight the role of design to the chosen context. Then, I focus on my own reflections and limitations during the study. Lastly, I highlight the opportunities for future work within this direction of speculative design for the kitchen.

6.1 Speculative design explorations in the field of HFI

6.1.1 Food and design

From hope to dream and addressing issues

Dunne and Raby (2014) mention that one of the key changes that has happened since the radical design movement of the 70s is that the dreams of the past were not sustainable, and now the younger generations no longer dream but hope, hope to survive and sustain themselves. Exploring societal issues through speculative design can take designers out of the conventional boundaries, to bring awareness and offer alternatives. It is also important for designers to acknowledge their power to create cultures with their designs (DiSalvo, 2021). Through the artifact analysis I brought forward the major changes that have happened to society. While there are many other factors influencing this process, artifacts reflect society and its values, or in other words the culture that they embody. This is why speculative design is so powerful, as it can bring artifacts embodying cultures that make people reflect on the status quo. With this thesis I endeavoured to bring awareness to our disconnection from food and possibly bridge that gap. I noticed that most awareness was caused during the workshop sessions when I actively invited the participants to imagine how they would survive in the scenario where food has to be sourced from scratch and cooked over fire. Their responses reflected their unpreparedness for such a future, stressing on the

importance of bringing awareness to the knowledge and skills we might be losing.

Preparation for future

One way to address the trend of efficiency is to think about the concept of Homo Ludens. To nurture creativity means to encourage exploration and mistakes. As I mentioned in the background chapter, one way to make cooking more pleasant is to make it challenging (Morozov, 2013), similar to a game. I began this thesis with Homo erectus and the domestication of fire, to highlight that approaching discomfort was the turning point in human evolution. Now, in the age of Homo Ludens, perhaps approaching the discomfort and making challenging artifacts is what people need in order to prepare for the future. Creativity, expression and mistakes should be embraced, and design is to support it, not defy the human nature. Future visions of smart technology are critiqued to have limited view on humans (Dunne & Raby, 2014). Hence, in a field filled with visions of aesthetics and future of smart technology, one design challenge stands up: who is in charge, technology or user (Bell et al., 2005)?

6.1.2 Sense of control

One of the major insights that came up from the testing of prototype 2 is related to the sensation of control with technology. The toastless toaster essentially was constructed to give people the impression of mastery, eventually revealing that their actions do not matter for the final result, but are mere superficial aesthetic element of the artifact. The theme of control emerged during the discussion with the participants, more specifically of the control over the black box that gives satisfaction, even though we do not understand it. As far as my research concerning SCD with focus on food practices, exploring the black box metaphor and the sense of control was under-developed opportunity. This artifact not only becomes a discussion tool, but also encourages people to “open” the black box and discover what is happening under, regaining that control. There is also the question of how you can mimic this feeling of mastery within the people. During a discussion with one of the participants they mentioned how it takes one successful attempt during troubleshooting to create this feeling:

> You poke around in things until you see a result. And sometimes you, even if it has no actual effect on something, you will still do it because one time it worked. Yes. Like how you bang on things to get it to work

It is interesting to explore further how such an approach may cause people to react and reflect on their current technology mediated lifestyle, very much based on this crave for control and mastery over things we do not understand.

What is important when evaluating speculative artifacts is that they do not imitate reality, but rather entertain ideas for alternative futures (Dunne & Raby, 2014). Being critical for this thesis, the scenarios as provotypes are more engaging in imagining alternatives worlds, compared to the prototypes themselves. Nevertheless, a physical representation of the speculation is vital in connecting it to the issue in practice. What brings depth to speculative artifacts, as highlighted by DiSalvo (2021) is engaging politics when addressing social issues. In this project, and especially with prototype 2, I focus on the politics of design, and how design can affect societal values, in an endeavour to cause reflection in both users and designers.

6.2 Critical reflections

In this subsection I focus on my personal reflections about the design process. I identified 3 main problem areas which I will go through in the following paragraphs.

6.2.1 Struggles with narrowing down ideas

A personal challenge that I can reflect on throughout this project is the fact that speculative design has a lot of freedom, which I could not manage well. A specific instance of this is the way I envisaged my opportunities after the fieldwork, illustrated in Figure 29. Initially, I recognised 3 opportunities worth designing for, by formulating different RQ for each. However, this was not sustainable as it pushed me to think in way too many directions, which is problematic in a space as ambiguous as the future. The future in design can be represented as a cone, as seen in Figure 30, exemplifying the vast opportunities.

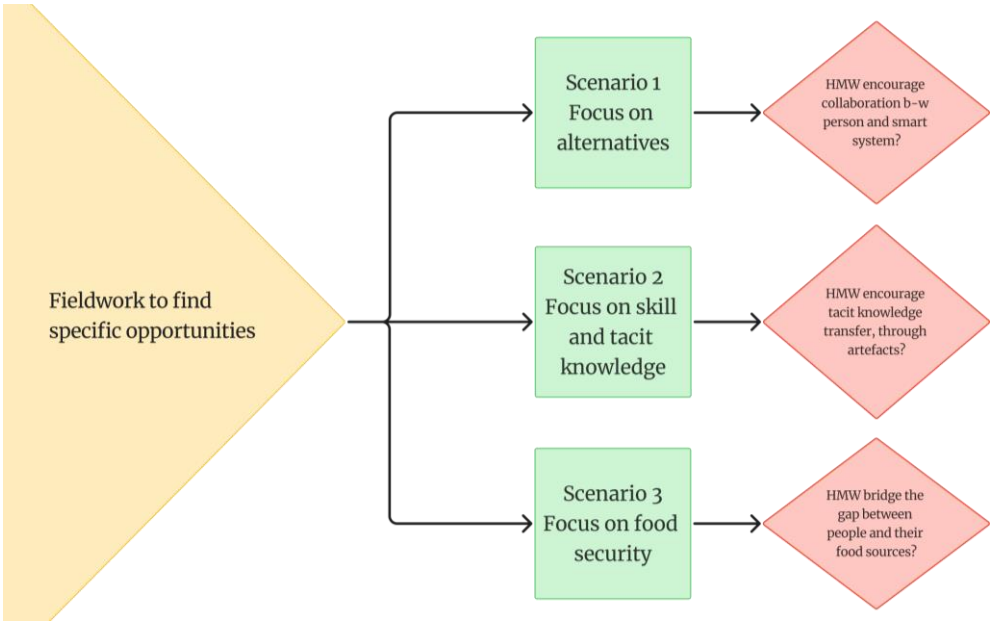


Figure 29. Initial view of the prototyping course

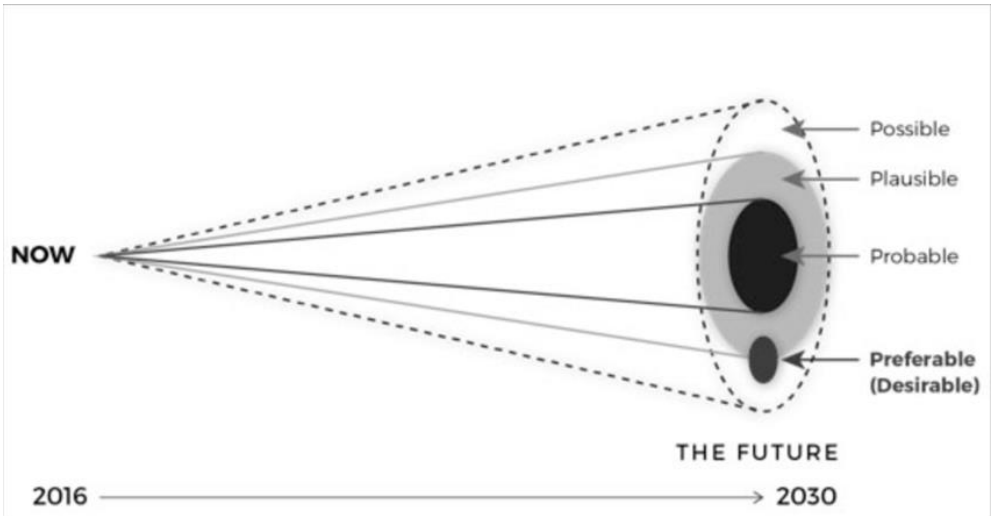


Figure 30. Adapted from the future cone [Image], by Hancock T., Bezold C., 1994, Europe PMC. (https://europepmc.org/article/med/10132155).

The way I coped with this was through building prototypes and essentially researching through design. This way, I could pick one opportunity and design within it. It affected my time schedule, which consequently led me to diverge more drastically. In this case it was useful, as I had begun exploring more things than I could have done alone.

6.2.2 Prototyping challenges

One major challenge I faced during this thesis was prototyping. As the weeks passed by, I noticed how my research was outweighing the design part of the project, essentially bringing imbalance. A good approach for SCD is rapid prototyping, in order to find the best way to convey one's message. However, prototyping was challenging for me, especially in making the prototypes functional, which took more time than what I expected in my schedule. Nevertheless, it also pushed me out of my comfort zone when working with materials and improvising. Some testing sessions were affected by the prototypes' fidelity, which also added to the mission of deciphering the meaning behind them, for better or worse.

6.2.3 Narrow testing group

Another important limitation of this study is the narrow testing group. I tested my prototypes with other design students out of convenience, hence most of the insights could be influenced because of their background. It is also important to consider that their food practices could differ from those of non-students.

Another implication with testing SCD artifacts is whether one should disclose their nature or not. I tested for both cases and it was interesting to see what differences in insights it would produce. Nevertheless, the testing groups were small so the insights are based on limited opinions.

6.1 Future research and design opportunities

As a speculative design project with a lot of freedom, it left many areas undiscovered. In this paragraph I will highlight the 2 paths that I did not take but see as worthy of exploring.

6.1.1 Designing for tool alteration to fit context

A common theme that I discovered during the fieldwork phase of the project was tool alteration. It came up in different shapes, not only through observation, but also through discussions during the workshops, primarily focused on re-using technology to fit the new lifestyle. A vital part of this theme is the situatedness and the context of the situation. Most participants acted upon what was already in front of them, adopting tools to their needs. It is interesting to explore speculative concepts through tools that can change depending on the context. However, an integral part of this idea is the human factor. It is by recognising the gap and connecting previous experiences, that it allows us to act upon it, similarly to the concept of coping (Heyer, 2018). This reflection can also be connected to the insight from testing prototype 2, where the participant connected the human factor to the sense of taste, as something machines cannot replace. Sutton (2006) also recognises taste as a skill that is highly neglected when thinking of cooking skills.

6.1.2 Connecting people to their food sources

An opportunity left under-discovered in this study is connecting people to their food sources. While my concept pushed people to think about where their food comes from, it did not really bridge that gap. An interesting concept that is worth exploring within this area is “food as a storyteller” (Bell & Kaye, 2002). It can be interpreted as exploring where the food came from, in terms of location, or how the recipe for something was passed down to us. In this way, not only can we connect people to their local food sources, but also connect people to people, such as in the case of the Malmö urban food revolution (STAD, 2024). One particular idea I had was hand-written cookbooks. A case of example is the cookbook that I always found in my grandma’s spice drawer, which held dozens of papers together in a charming chaos. Things that always strike me when perusing it are the names of the recipes, connecting each recipe to the person that shared it with her. The stains on the sheets also show how the book has been present in a lot cooking sessions, holding the smells of ingredients that have fallen on it. Lastly, it is also her handwriting and measurements that bring me closer to her interpreting the recipe, something that goes against standardization.

7 Conclusion

> "...it is just funny how you said we conquered the fire long time ago and now we cannot even make it."

A reflection from Participant 5 in one of the testing sessions confirmed the overall unpreparedness of people for the future catastrophes. While the aim of this thesis was not to teach people how to make fire, it aimed at designing exaggerated kitchen artifacts, a synecdoche of possible future catastrophes. The main perpetrator in focus is food security and efficiency in kitchens. I began my project with academic research on what is already there, including past designs of the stove and comparing it to the present, and offering a view of the "future stove" and its values. I used this knowledge to create future scenarios which I used together with participants in workshops to spark their creativity in brainstorming sessions. Before entering the materialisation phase, I pinpointed three main themes worth designing around, which also helped me in tackling the challenge of narrowing down ideas. As a result, I created two prototypes. The Preservation Scale, which is a discussion tool around food's longevity and preservation methods, that allowed people to evaluate their knowledge and bring out traditions from their past. And the Toastless Toaster, which is an endeavour to critique the future of over-engineered technology, by giving the user the illusion that they control what is happening in the black box, only to be revealed otherwise. I concluded the study by building on past SCD theory surrounding food practices and exploring the gaps of interaction design and food cultures. An important concept that I propose through my design is the mastery of the black box. In other words, giving people the idea of control of the technology, just to reveal it is acting on its own. As such, I bring attention to the overarching question of who is in control.

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9 Appendix

9.1 Information letter and Consent form

MALMÖ UNIVERSITY

Information letter	Appendix 1

Project title: Reimagining Kitchen Artifacts through Future Catastrophes	**Date: 01.04.2025**
Study manager: Lora Yordanova **E-mail: lorayo2003@gmail.com**	**Studying at Malmö University,** **Faculty of Culture and Society,** **S-205 06 Malmö,** **Phone: +46 40 665 70 00** **Education: Interaction Design** **Level: Bachelor Degree**

My name is Lora Yordanova and I am studying Interaction Design at Malmö University. It is my final semester and I am currently doing my bachelor thesis: Reimagining kitchen artifacts through future catastrophes. The thesis is carried out by me, and supervised by one of my professors.

The aim of my study is to critique the current focus of design for the kitchen, which is primarily influenced by efficiency. With my research I want to investigate how design currently affects food practices, which will be used to create future scenarios and prototypes, to encourage reflection on the present.

The study will be carried out in 2 phases. First, I will conduct fieldwork with focus on food practices and how current design affects them. This consists of observations, interviews and brainstorming sessions. Second, after I have created prototypes I will test them, utilizing similar methods. I will employ as many participants as possible, selected by convenience of both parties. Each study session will not exceed 2 hours and will be accompanied with breaks and snacks.

Your participation in this study is entirely voluntary. You are free to quit the study at any time, and are not required to give any explanation.

I strive to protect any personal data and information that may surface during the study, from any unauthorized people who may have access to the material. The material is stored in my personal storage space in Malmö University and is only accessible by me. In the final report of my thesis, respondents will remain anonymous, making it not possible to trace the result to the real identities of the participants.

My thesis will be submitted to K3, for evaluation and presented to small audience during my defense. The detailed work will be published in the DIVA Portal, where all student theses are published.
The link to this is: https://mau.diva-portal.org/smash/search.jsf?language=sv&dswid=7897

You are hereby asked to take part in this study.

MALMÖ UNIVERSITY

Informed consent Appendix 2

Project title: Reimagining Kitchen Artifacts through Future Catastrophes	**Date: 01.04.2025**
Study manager: Lora Yordanova **E-mail: lorayo2003@gmail.com**	**Studying at Malmö University,** **Faculty of Culture and Society,** **S-205 06 Malmö** **Phone +46 40 665 70 00** **Education: Interaction Design** **Level: Bachelor**

I have been verbally informed about the study and read the accompanying written information. I am aware that my participation is voluntary, and that I, at any time and without explanation, can withdraw my participation. The person/-s leading the study will strive to guarantee confidentiality in that no unauthorized person may have access to the material. The gathered material will be stored properly and used for research purposes only.

I hereby submit my consent to participate in the above survey:

Date: ……..

Participant's signature: ……

9.2 Scenario #1

Scenario 1

The Dystopian Future of The Smart Kitchen

The year is 2150 and the kitchen is located outside of the house. The thing connecting the kitchen and the home is a conveyer belt that brings the ready food and takes the dirty dishes. The corporations have achieved the absolute automation ideal, allowing people, especially house wives to pursue the things they wish to.

Let's look into detail of how the things actually work. We have our revolutionary Specimen A who want to make kanellbullar from scratch to remember his childhood. We could say this is his comfort food, so he wants to make it like they used to in the last century.

When he enters the kitchen he is met with a very futuristic factory-like layout. The kitchen colours are white and minimalistic, everything is touch screen and lacks affordances, because machines don't need that. The devices communicate between each other through signals that A cannot perceive, but sometimes notices. In a sense the kitchen is its own ecology of machines that works as one to produce the desired product.

To begin the process he has to put on the APPron, the apron from the future, equipped with everything you need to make cooking efficient process. It is essential to wear it, for it allows you to actually interact with the rest of the kitchen, as a fellow machine. If you do not wear the Appron then the kitchen will think you are a virus and try to eject you out of it.

A puts on the Appron and looks around. What he sees first is a white counter top where he decides to start. He tries to open the cabinet above but fails to. Then he receives a signal from the Appron that he needs to use the exoskeleton to open the cabinets, so he enables it. A skinny metal arms emerges from his back opening the cabinet and exposing the inside, an almost never ending conveyer of bowls. The arm places the bowl on the counter and A starts looking around for the fridge. He goes to the fridge and when he opens it he is met with an unbelievable view. The fridge is a small farm that grows the food inside of itself. While trying to source the products A accidentally disturbs the ecology of the fridge and suddenly the things start to rumble. He quickly closes it while holding milk, eggs and salami which he quickly gathered. The Appron tells him that his hands are too impure to touch the fridge without consequences. He places the products on the counter but soon realises that he didn't need salami but butter. The Appron assures him that this wont be a problem for the Teleport-wave. Puzzled he turns around and sees an opened microwave. As he approaches the Appron tells him that if he puts the salami inside he could exchange it for butter. Essentially, you sacrifice the salami for butter. He does this and starts the microwave on the highest setting. After 10 seconds he pulls out what seems like a piece of meat plasma. The Appron says you need to heat it for 10 seconds more. He does so and then pulls out the butter. He thinks what kind of magic is this.

As he has everything he starts putting the ingredients in the bowl, while the Appron is giving him directions and the arms are trying to do as much as possible. At some point A is annoyed and irritated expresses that he can do it himself. The Appron says that it knows the best method and to listen to it. A wants to add a touch of cardamom because that's what his family used to do before but the system does not approve, because this is not scientifically approved method. They go back and forth but A does not succeed. The robotic arms also do the kneading, as to make him feel as if he is kneading. A just watches and tries to interfere to feel the dough to no avail.

As he is about to put the buns to ferment the system says that fermentation happens while they bake in the quantum oven. Puzzled, A says that that's not how yeast works. Appron says that the oven has the quality to do everything in an instant. As A approaches the oven he sees it cannot be opened. That's when the Appron gives a signal to the oven and it opens. It looks rather different. As A places the tray the oven closes. He goes to the sink to bring the dirty dishes. He sees that the garbage disposal is full and is working. As he waits for it, he notices that the sink has produced a lunch box out of the scraps. The Appron explains that it is a Zero-waste sink and that it uses everything that goes to waste. Disgusted he looks at the lunch box and sees that it looks rather appealing. In this moment the oven opens and present the ready buns. The Appron says that they are ready and A can go out so they kitchen can clean itself. A is rushed out of the kitchen with a box of cinnamon buns.

9.3 Scenario #2

Scenario 2

The catastrophic future around food security

The year is 2100 and the world is hit by a massive disaster called food security. As companies were focused so much on efficiency and technological innovation all resources were drained of the earth. This has forced all logistic companies to bankrupt and hence all transportation has ceased. As people were so influenced by the consumer culture and search for the newest, the bigger and the better they overlooked one of the most important factors of their survival - food. Until now food was just something they had to consume out of pleasure and gluttony was growing with consumerism. Everything was accessible in the supermarket, from frozen fast meals to exotically sourced delicacies and fruits. Now the supermarket is barren, with only number of things nobody wants left.

Humans were forced to adjust their lifestyle. Their values moved from consumerism to survival. The first thing they did was turn to the internet and try to relate to other people. However, they soon found out that signal has ceased as well. Electricity was next, they thought. This very apocalyptic picture has developed in front of people's eyes so quickly. Immediately, they came out of their houses to look for their neighbours, the first time they have probably met them. Lets focus on Specimen B and his journey through this new environment.

B came out of his house and looked around for his neighbours. They were doing the same. As they gathered they started conversing, expressing how ironic it is to finally meet in such circumstances. People started gathering in groups depending on the size of their neighbourhood and soon started brainstorming ways of surviving.

What everyone had to deal with is go back to basics. However, not everyone was an expert camper, who knows survival tricks, someone said. The response of another person was that they all have strengths and they have roles in their small society. This person soon became a leader of the group. But let's focus on even smaller unit, our specimen B.

B is lucky as he lives in a house, which has way more resources than a conventional apartment. B has a big open kitchen that overlooks his garden. To survive he has to obtain food. Unfortunately no electronics are working to get a 101 tutorial as to how to do agriculture or hunt. B was puzzled, all his technology suddenly betrayed him. He understood how reliant he was on it. Everything he did was mediated through technology. So he needed to adapt.

Thankfully, he still held onto his great-great grandfather's diary from his time when they lived in the far-away cottage. The diary was worn out and had stains on its pages, omitting a smell of an old house, wood and nature. Like a book that has travelled through time it was an interesting artefact for B. He has read it with his father and conversed over stories from this great grandfather. He loved doing things from scratch.

What B started with was a plan to hunt his dinner and to begin farming. Because he lived in the suburbs he was close to the forest and nature. He soon realised that hunting was not sustainable for him and the community. So it was decided that people will start farming animals. Everyone was delegated some animals. As it happens B was taking care of cows. Surprisingly he didn't know that butter came from cows. He was that alienated from his food in the past, only ordering fast food or frozen meals. He also didn't know that the cow needed to have a baby to produce the milk, which further puzzled him. What was the most shocking to him was the difficulty of agriculture and planting. It took him so many tries to plant cucumbers and tomatoes that he was close to giving up.

After a few months he realised that winter is coming, and his vegetables cannot survive. He soon found himself consulting with the book about preservation of food. He was surprised to learn how to preserve food, through pickling, dry-aging and vacuuming. Of course, during winter he could use outside as a fridge, but he also needed to preserve food for his cows. Learning how to make fire inside of his home was also important. However, his kitchen cannot support his new lifestyle. The oven doesn't work and it cannot hold fire inside. The water supplies will soon run out as well, so the sink is just raw materials. His electronic gadgets cannot serve any purpose, but be recycled into parts.

He turned his sink into a garden, where he could dispose of old food to transform into compost and to grow vegies in the winter. He created new equipment for his cooking, such as spatulas, mixer and blender that works by hand. His survival instinct drove his creativity in re-creating new things, to meet his needs. The kitchen became the place where he would turn nature into culture. With the time he had, he also explored old recipes, trying to remember comfort foods from his childhood.